

CASE STUDY:

Digital Transformation & Enterprise Systems at Universiti Tun Hussein Onn Malaysia (UTHM)

(For Education Purposes – Based on publicly available information)

1.0 Introduction

Universiti Tun Hussein Onn Malaysia (UTHM), located in Batu Pahat, Johor, is a public technical university known for its strong engineering, technology, and vocational education focus. With approximately **18,000 students** and a diverse community of academic and administrative staff, UTHM operates a complex ecosystem of academic, residential, administrative, and student-services processes.

As the university grows, the need for **integrated, efficient, and user-friendly digital systems** becomes increasingly critical. UTHM has taken significant steps toward digitalisation through its **Student Information System (SMP)**, **Staff Information System (SMS)**, the **UTHM SSO portal**, and its official **MyUTHM mobile application**, which provides access to selected campus services. However, many operational areas still function in silos or rely on fragmented systems, leading to data duplication, inconsistent workflows, and manual interventions.

This case study provides a realistic environment for students to analyse UTHM's digital architecture and design improved enterprise solutions using **SAP Build Apps**.

2.0 Existing Digital Systems at UTHM

Based on publicly available information, UTHM currently operates several core systems supporting its academic and administrative operations.

2.1 Student Information System (SMP – Sistem Maklumat Pelajar)

SMP is the primary system used for managing the student lifecycle:

- Student biodata
- Programme registration & course enrolment
- Examination records & results
- Academic status tracking
- Residential allocation links (selected functions)

This system forms the backbone of UTHM's academic administration.

2.2 Staff Information System (SMS - Sistem Maklumat Staf)

SMS supports:

- Staff biodata & profiles
- Leave records
- Service history
- Workload management (limited)
- Employment documentation

2.3 UTHM Single Sign-On (SSO) Portal

UTHM uses a unified identity management system that allows students and staff to log in once to access multiple services. The SSO serves as a **central authentication gateway**, improving user convenience and security across systems.

2.4 MyUTHM Mobile App

The MyUTHM app provides selected digital services depending on the user's role:

For Students

- Viewing biodata
- Virtual student ID (QR code)
- Shuttle bus live tracking
- Academic calendar
- Quick links to LMS, examination info
- Event listing & reminders

For Staff

- Attendance & leave
- Activity dashboard
- Personal profile
- Internal updates

Despite these features, many student-facing processes (hostel, clubs, bookings, advising, assessments) remain **unintegrated or handled manually/outside the app**.

2.5 Other Known or Expected Systems

Although not fully documented publicly, typical university systems likely exist:

- Hostel/Residential Management System
- Finance & e-payment system
- Library information system
- Research & grant management
- Asset management (labs, equipment)
- IT helpdesk / maintenance ticketing

These systems may exist as standalone solutions or partially linked to SMP/SMS.

3.0 Key Challenges faced by UTHM's Digital Ecosystem

Based on typical public-sector university environments and observable gaps:

3.1 System Fragmentation

Multiple systems exist, but they do not fully integrate.

Example: hostel data may not automatically update student status in SMP.

3.2 Manual Processes Still Prevalent

Some processes — academic advising, club management, maintenance requests — remain paper-based or rely on Google Forms/Excel.

3.3 Data Silos

Data is duplicated across systems (e.g., student details stored in apps, hostel database, and faculty records).

3.4 Limited Mobile Integration

While MyUTHM exists, not all services are digitised, and many processes require students to access different portals.

3.5 Lack of Real-Time Analytics

Operational decisions (room allocation, student progress monitoring, critical students activities information) lack real-time dashboards.

4.0 Opportunity for Digital Enhancement

UTHM's direction toward a **Smart Campus** positions it well for improving:

- Student services
- Campus operations
- Academic administration
- Transport & facility management
- Digital workflows & approvals

The existing MyUTHM app can be strengthened by integrating more backend services and providing a unified student experience.

This presents an excellent opportunity for students to **design realistic, relevant, and impactful subsystems**.

5.0 Subsystems for Group Project Development

For this project, each group will develop **ONE subsystem** using **SAP Build Apps**. Your subsystem should be designed as an enhancement or extension of **UTHM's existing digital ecosystem**, focusing on solving real operational problems or improving the user experience.

Each subsystem must clearly demonstrate:

1. **Functional Requirements**
2. **User Interface (UI) Design**
3. **Data Model (entities, relationships, attributes)**
4. **Workflow Logic (process steps, CRUD operations, approvals, notifications)**
5. **Integration Points** with assumed UTHM master data (e.g., student, staff, facility records)

You are expected to analyse the current situation, identify gaps, and propose a modern digital subsystem prototype that could realistically fit into UTHM's environment.

5.1 Academic Subsystems

(Enhancing SMP-related academic processes and student experience)

A. Subject Registration Enhancement Module

This subsystem improves the course registration process and may include:

- Course browsing and filtering
- Timetable conflict detection

- Credit hour validation
- Digital Add/Drop request and approval workflow

B. Academic Advising & Progress Tracker

This subsystem supports students and advisors through:

- Appointment scheduling with academic advisors
- Recording advising notes and recommendations
- Viewing academic progress and performance trends
- Early-warning alerts for at-risk students

5.2 Campus Life Subsystems

(Supporting student residential and extracurricular needs)

C. Hostel Management System

This subsystem focuses on managing student accommodation:

- Hostel room application and allocation
- Check-in and check-out processes
- Maintenance request submission and tracking
- Hostel fee monitoring

D. Student Clubs & Activities Management

This subsystem aims to support extracurricular engagement:

- Club profiles and membership records
- Event creation and student registration
- Attendance tracking for activities
- Borrowing and returning club assets/equipment

5.3 Operational & Administrative Subsystem

(Supporting UTHM's non-academic operations)

E. Campus Transport Schedule & Shuttle Tracking

This subsystem improves transport services by offering:

- Bus route displays and schedules
- Live shuttle location (simulated for the project)
- Estimated arrival times
- Feedback or complaint reporting

GROUP 1: Subject Registration Enhancement Module

(5 members → 5 sub-modules)

Member	Sub-Module	Description
1	Course Catalogue Browser	List/search courses, filter by programme, semester, credit hours.
2	Timetable Conflict Checker	Validate selected courses against student timetable; highlight clashes.
3	Credit Hour Validator	Calculate total registered credit hours; apply min/max rules; warning alerts.
4	Add/Drop Application Form	Student submits add/drop request with reasons; UI for request creation.
5	Add/Drop Approval Workflow	Advisor/Faculty view requests, approve/reject, add comments, notify student.

Integration assumption: Student profile → from SMP master data.

GROUP 2 : Academic Advising & Progress Tracker

(6 members → 6 sub-modules)

Member	Sub-Module	Description
1	Advising Appointment Scheduling	Students book advising sessions; advisors set available slots.
2	Advising Notes & Recommendations	Advisors write notes, attach recommendations, view history.
3	Academic Progress Dashboard	Displays CGPA, GPA, completed credits, progress charts.
4	Course Performance Viewer	Shows detailed performance for each subject (CA, exams, attendance).
5	Early Warning Alert System	Flags at-risk students based on rules (low GPA, poor attendance).
6	Student Advising Records & Document Uploads	Upload & store documents (study plan, appeal letters), linked to advising history.

Integration assumption: SMP → academic records, attendance, CGPA data.

GROUP 3: Hostel Management System

(5 members → 5 sub-modules)

Member	Sub-Module	Description
1	Room Application Module	Students apply for hostel; select room preferences; view application status.
2	Room Allocation Module	Admin allocates rooms; view availability; manage room assignment.
3	Check-In / Check-Out Module	Digital check-in/out forms; room condition checklist; QR-based verification.
4	Maintenance Request Module	Students submit maintenance tickets; admin assigns technician; track status.
5	Hostel Fee Monitoring Module	Display hostel charges, payment status, reminders; link to payment system.

Integration assumption: Student active status → SMP. Payment → Finance system.

GROUP 4 : Student Clubs & Activities Management System

(6 members → 6 sub-modules)

Member	Sub-Module	Description
1	Club Profiles & Membership Management	Club listings, join/exit club, member roles (President, Treasurer).
2	Event Proposal & Approval Workflow	Committee proposes event; advisor approves/rejects digitally.
3	Event Registration + QR Code Generation	Students register; unique QR code generated for check-in.
4	Event Attendance Scanning & Tracking	QR code scanning; attendance stored; summary page.
5	Asset Borrowing System for Club Events	Borrow equipment (PA system, chairs, laptops); approval + return logging.
6	Club Finance & Budget Tracking	Records event budgets, club expenses, fund usage (simple ledger).

Integration assumption: Staff advisor list → SMS. Facility booking → Facility system.

GROUP 5 : Campus Transport & Shuttle Management

(6 members → 6 sub-modules)

Member	Sub-Module	Description
1	Route & Schedule Display Module	Displays bus routes, operating hours, stops, and campus map.
2	Live Shuttle Tracking (Simulated GPS)	Simulated real-time shuttle position via periodic API or variable updates.
3	Estimated Arrival Time (ETA) Predictor	Shows countdown timers for next bus arrival at selected stops.
4	Shuttle Occupancy Monitoring (Simulated)	Simulated occupancy level (e.g., low/medium/high) to help students plan.
5	User Feedback & Complaint Reporting	Students submit feedback/issues; admin reviews & updates resolution status.
6	Transport Analytics Dashboard	Visual charts for admin: peak hours, route usage, complaints received.

Integration assumption: Student profile → SMP; contact details → MyUTHM profile.

6.0 Integration Scenario (Enterprise Architecture Context)

To simulate a real university digital ecosystem, assume that UTHM already operates several centralised data sources and enterprise services:

- **Master Student Table** (from SMP)
- **Master Staff Table** (from SMS)
- **UTHM Single Sign-On (SSO)** for authentication

Your subsystem should **conceptually integrate** with these existing components by:

Example Integration Points

- A **Hostel Management System** reads *student status* from SMP (active, defer, etc.)
- An **Advising System** retrieves *academic performance data* from SMP
- A **Shuttle Tracking System** displays personalised information based on *user profile* from MyUTHM

Your SAP Build prototype does not need real integration, but you must **design and explain how integration would logically occur**. This ensures your project aligns with Enterprise Architecture principles while maintaining a manageable development scope.